

Higher Order Effects

Agenda

- 1) Review Estimates
- 2) Causal Parameters
- 3) 3 Factor Design
 - ↳ Making Paper

The F-statistic

"The variability of the group averages
relative to the
average variability within each group."

$$\frac{A}{B} = \frac{MS_B}{MS_w} = \frac{\frac{1}{J-1} \sum_{j=1}^J n_j (\hat{\bar{Y}}_j - \hat{\bar{Y}})^2}{\frac{1}{n-J} \sum_{i=1}^n (Y_i - \hat{\bar{Y}}_{j(i)})^2}$$

Estimates

$$Y_i = \mu + \alpha_{j(i)} + \beta_{k(i)} + \gamma_{j(i)k(i)} + \varepsilon_i$$

$$\hat{\mu} = \hat{\bar{Y}}$$

$$\hat{\alpha}_j = \hat{\bar{Y}}_j - \hat{\bar{Y}}$$

$$\hat{\beta}_k = \hat{\bar{Y}}_k - \hat{\bar{Y}}$$

$$\gamma_{j,k} = \hat{\bar{Y}}_{j,k} - \hat{\alpha}_j - \hat{\beta}_k - \hat{\mu}$$

$$Y_i - \hat{Y}_i = Y_i - (\hat{\mu} + \hat{\alpha}_{j(i)} + \hat{\beta}_{k(i)})$$

$$F_A = \frac{MS_A}{MS_R} = \frac{\frac{1}{J-1} \sum_{j=1}^J n_j \alpha_j^2}{\frac{1}{n-1-(J-1)-(K-1)} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2}$$

If $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$

then under $H_0: \alpha_1 = \alpha_2 = \dots = \alpha_J = 0$

$$F_A \sim F(df1 = J-1, df2 = n-1-(J-1)-(K-1))$$

Causal Parameters

[Slides]

Model for 3 factors w/ interactions

$$Y_i =$$

[Slides]